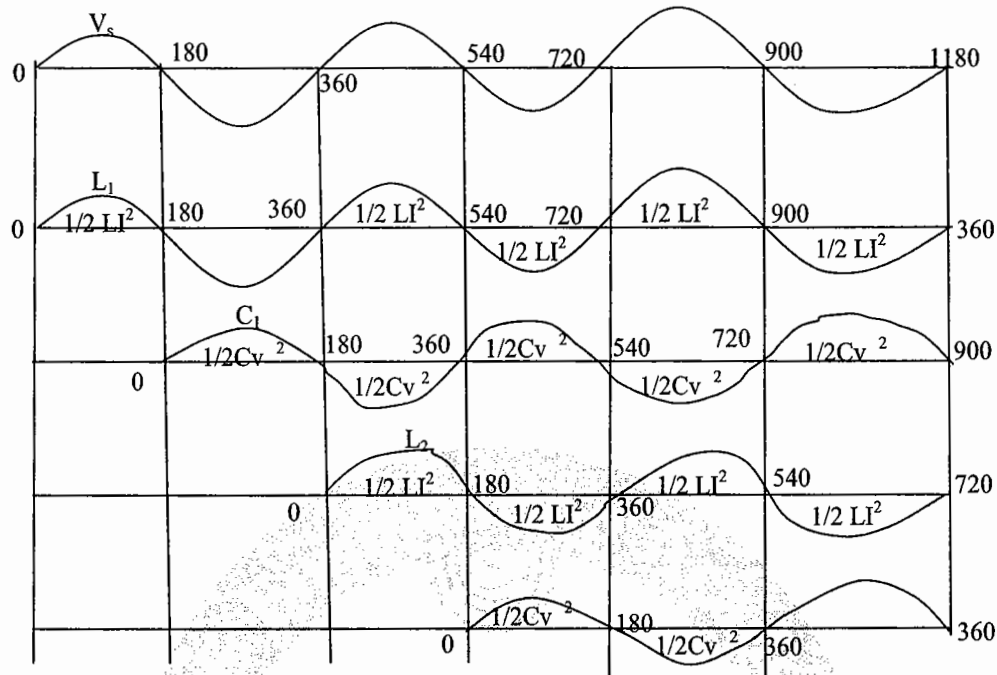
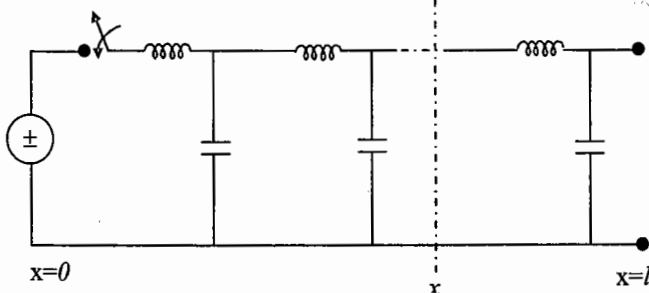




When the inductor is charging from 360-540° it will discharge the energy of 180-360° to capacitor so that the capacitor will charge from 180-360°. But the energy stored in capacitor from 0-180° will deliver to next inductor to develop the magnetic field from 0-180°. The process is continuous and the wave is traveled from one end to the other end is by gradually building up of the magnetic field and the static field. When the wave reaches the load end, it will be opposed by load impedance so that it will offer a reflection for the incident wave due to change in the medium.

Velocity of wave propagation :



Inductance per km is L , capacitance per km is C .

Total magnetic flux linkages upto x km is

$$\psi_x = L I \cdot x$$

Voltage in the line $V = \frac{d\psi_x}{dt} = LI v$ ---- (1)

Where v is velocity of wave propagation

Total electric flux linkages upto x km is

$$\lambda_x = C V x$$

Current in the line $I = \frac{d\lambda_x}{dt} = CV v$ ---- (2)

Multiply equations (1) & (2)

$$VI = (LC) (VI) v^2$$

$$v^2 = \frac{1}{LC}$$

$$v = \frac{1}{\sqrt{LC}} \text{ m/s}$$

Where $L = \frac{\mu_0}{2\pi} \ln \left(\frac{GMD}{GMR} \right) \text{ H/m}$

$$C = \frac{2\pi\epsilon_0\epsilon_r}{\ln \left(\frac{GMD}{r} \right)} \text{ F/m}$$

$$v = \frac{1}{\sqrt{\frac{\mu_0}{2\pi} \ln \left(\frac{GMD}{GMR} \right) \frac{2\pi\epsilon_0\epsilon_r}{\ln \left(\frac{GMD}{r} \right)}}}$$

$$\text{generally } \ln \left(\frac{GMD}{GMR} \right) - \ln \left(\frac{GMD}{r} \right)$$



$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0 \epsilon_r}}$$

- For over head line system, surroundings dielectric medium air $\epsilon_r = 1$ for air

$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \frac{1}{\sqrt{4\pi \times 10^{-7} \times 8.854 \times 10^{-12}}}$$

$$v = 3 \times 10^8 \text{ m/s}$$

$$v = 3 \times 10^5 \text{ km/s (for lossless OH line)}$$

- For attenuated over head line,
 $v = 2.5 \times 10^5 \text{ km/s to } 2.8 \times 10^5 \text{ km/s}$

- For Underground cable

$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0 \epsilon_r}}$$

$\epsilon_r > 1$ for cable dielectric material

$$v = \frac{3 \times 10^8}{\sqrt{\epsilon_r}} \text{ m/s}$$

$$v(\text{UG cable}) > v(\text{OH line})$$

Example 2.4.1:

What is the velocity of wave propagation in a under ground cable having relative permittivity of its dielectric material 4.

Sol: in UG cable $v = \frac{3 \times 10^8}{\sqrt{\epsilon_r}} \text{ m/s}$

Where $\epsilon_r = 4$. so, $v = 1.5 \times 10^8 \text{ m/s}$

Example 2.4.2:

The time taken for a surge to travel a 600 km long overhead transmission line is (IAS-07)

- (a) 6 s (b) 1 s
(c) 0.02 s (d) 0.002 s

Sol: The time taken by the wave to reach end of

the line is $T = \frac{\text{length}}{\text{Velocity}}$

$$T = \frac{600}{3 \times 10^5} = 0.002 \text{ s}$$

Propagation constant (γ) :

It represents changes happening in V,I waves during wave propagation such as physical displacement and decrement in magnitude of wave.

$$\gamma = \sqrt{zy}$$

$$z = r + j\omega L \Omega / \text{km}$$

$$y = g + j\omega C \text{ U/km}$$

- For attenuated Tr line

$$\gamma = \sqrt{(r + j\omega L)(g + j\omega C)}$$

$$\gamma = \alpha + j\beta$$

α is attenuation constant, represents decrement is V,I wave magnitude (nepers/ km)

β is quadrature component, represents physical displacement of wave (radians/ km)

- For loss less line ($r = 0$, $g = 0$)

$$\gamma = j\omega \sqrt{LC}$$

$$\alpha = 0, \text{ no attenuation}$$

$$\beta = \omega \sqrt{LC} \text{ rad / km}$$

$$\beta = 1.047 \times 10^{-3} \text{ rad / km}$$

- For loss line or attenuated line

$$\beta > 1.047 \times 10^{-3} \text{ rad / km}$$

Wave length of line (λ) :

Wave length of transmission line is distance to be traveled by the wave to get one wave repetition or 2π radians or 360°

With β rad/ km physical displacement

$$\lambda = \frac{2\pi}{\beta}$$

Example 2.4.3:

The propagation constant of a transmission line is:

$$0.15 \times 10^{-3} + j1.5 \times 10^{-3}$$

The wave length of the traveling wave is

(a) $\frac{15 \times 10^{-3}}{2\pi}$

(b) $\frac{2\pi}{15 \times 10^{-3}}$

(c) $\frac{15 \times 10^{-3}}{\pi}$

(d) $\frac{\pi}{15 \times 10^{-3}}$



Sol: If the propagation constant is $\alpha + j\beta$ then the wave length will be

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{15 \times 10^{-3}} \text{ km}$$

Surge impedance / characteristic impedance (Z_c) Natural impedance :

It is impedance offered by apparatus for surges or traveling waves or transients .

$$Z_c = \sqrt{\frac{Z}{Y}}$$

- For attenuated line ($r \neq 0, g \neq 0$)

$$Z_c = \sqrt{\frac{r + j\omega L}{g + j\omega C}}$$

$$Z_c = |Z_c| \angle -\theta \text{ (capacitive)}$$

- For loss less line: ($r = 0, g = 0$)

$$Z_c = \sqrt{\frac{L}{C}} \text{ (pure resistive in nature)}$$

It is pure resistance in nature but it will not create any losses in line.

Surge impedance or different apparatus

- For OH line $Z_c = 200 - 500 \Omega$

Typically for OH line $Z_c = 400 \Omega$

- For UG cable $Z_c = 40 - 80 \Omega$

Typically for UG cable $Z_c = 40 \Omega$

Example 2.4.4:

The Generalised circuit constants of a 132kV line are $A = D = 0.98 \angle 0.3^\circ$

$$B = 82.5 \angle 76^\circ \Omega$$

$$C = 0.0005 \angle 90^\circ \text{ mho}$$

What is surge impedance of line

- (a) $400 \angle 0^\circ$ (b) $400 \angle -7^\circ$
(c) $406.2 \angle -7^\circ$ (d) $300 \angle 0^\circ$

Sol: $Z_c = \sqrt{\frac{A}{C} \frac{B}{D}} = \sqrt{\frac{B}{C}} \quad (\because A = D)$

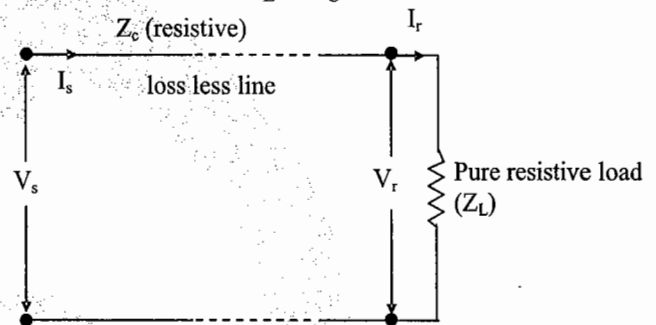
$$= \sqrt{\frac{82.5 \angle 76^\circ}{0.0005 \angle 90^\circ}} = 406.2 \angle -7^\circ \Omega$$

Surge Impedance Loading (SIL):

It is also known as characteristic impedance loading or ideal loading or ideal power transfer capability .

It is the loading on the line when ever the load impedance is exactly matching to surge impedance of transmission line .

So at SIL condition $Z_L = Z_c$



Loading = SIL

- $Z_L = Z_c$
- $\frac{1}{2} LI^2 = \frac{1}{2} CV^2$
- Transmission line will act as neither sink nor source of reactive power (Q)
- $|V_r| = |V_s|$
- $|I_r| = |I_s|$
- Load angle $\delta = \beta \ell$
- Both sending end and receiving end power factors are unity .

Loading > SIL

- $Z_L < Z_c$
- $\frac{1}{2} LI^2 > \frac{1}{2} CV^2$
- Transmission line acts as sink of reactive power
- $|V_r| < |V_s|$
- $|I_r| > |I_s|$
- Load angle $\delta > \beta \ell$



- Receiving end power factor is unity but sending end power factor is lagging

Loading < SIL

- $Z_L > Z_C$
- $\frac{1}{2}LI^2 < \frac{1}{2}CV^2$
- Transmission line acts as source of reactive power
- $|V_r| > |V_s|$ (ferranti effect)
- $|I_r| < |I_s|$
- Load angle $\delta < \beta\ell$
- Receiving end power factor is unity but sending end power factor is leading.

Economical loading on overhead line is loading > SIL.

Economical loading on under ground cable is loading < SIL.

Example 2.4.5:

A 800 kV transmission line is having line inductance and capacitance/ phase per km are 1.1 mH, 11.68 nF. Ignoring the length of line, ideal power transfer capacity in MW.

$$\begin{aligned}\text{Sol: Surge Impedence, } Z_C &= \sqrt{\frac{L}{C}} \\ &= \sqrt{\frac{1.1\text{m}}{11.68\text{nF}}} \\ &= 306.9 \Omega\end{aligned}$$

$$\text{Ideal power transfer capability (or) SIL} = \frac{V^2}{Z_C}$$

$$\begin{aligned}\text{SIL} &= \frac{800^2}{306.9} \text{ MW} \\ &= 2085.4 \text{ MW.}\end{aligned}$$

Example 2.4.6:

A 200km long lossless overhead line operated at SIL condition. What will be the load angle of transmission line _____?

$$\begin{aligned}\text{Sol: at SIL condition, load angle } \delta &= \beta\ell \text{ radians} \\ \delta &= 1.047 \times 10^{-3} \times 200 \\ &= 0.2094 \text{ radians} \\ &= 12^\circ\end{aligned}$$

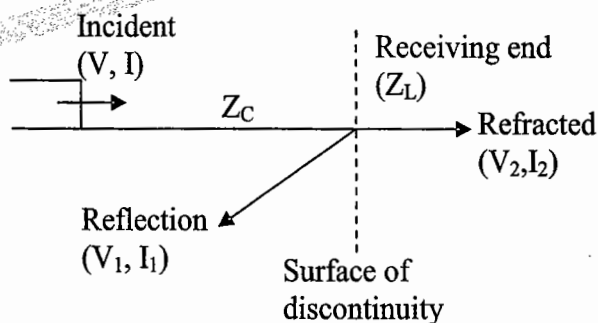
2.5 Wave Travelling Analysis

Whenever the traveling wave reaches to a discontinuity or if the wave faces a change in medium in its propagation then there will be some reflections experienced by the wave such that the resultant wave at that point will be sum of incident and reflected waves.

From basis wave equation, voltage at any point x, at any time t is given as

Where f_1 is forward wave and f_2 is backward wave.

Refracted wave = Incident wave + reflected wave



If incident voltage is V then the incident current

$$\text{will be } I = \frac{V}{Z_C}.$$

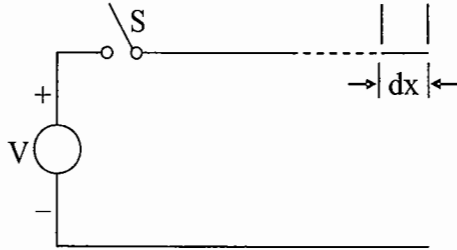
Refracted quantities are,



$$V_2 = V + V_1$$

$$I_2 = I + I_1$$

Open circuit line wave propagation



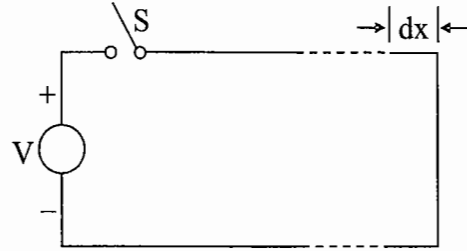
A step voltage of magnitude V initiated on transmission line $t=0$ instant such that a current wave of I also propagated towards end of the line.

In open circuit line always refracted current is zero at receiving end side where as voltage at sending end side is maintained as constant at V . Time taken by the wave for one propagation through the line is $T = \text{length} / \text{velocity}$.

Voltage	Current
$t=0$	
$t=T$	
$t=T^+$	
$t=2T$	
$t=2T^+$	
$t=3T$	
$t=3T^+$	
$t=4T$	

After four propagations the first propagation will be repeated.

Short circuit line wave propagation



A step voltage of magnitude V initiated on transmission line $t=0$ instant such that a current wave of I also propagated towards end of the line.

In short circuit line always refracted voltage is zero at receiving end side where as voltage at sending end side is maintained as constant at V . Time taken by the wave for one propagation through the line is $T = \text{length} / \text{velocity}$.

Voltage	Current
$t=0^+$	
$t=T$	
$t=T^+$	
$t=2T$	
$t=2T^+$	
$t=3T$	
$t=3T^+$	
$t=4T$	



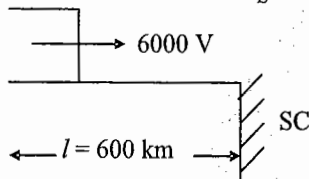
In each propagation the current in line will be increased by a value I such that in the fourth propagation the current in line becomes $4I$.

Example 2.5.1:

An overhead line 600 km long and having surge impedance at 500Ω . The line short circuited at the receiving end and a voltage of 6000 v is applied to the other end. Calculate the sending end voltage and current after 0.008

Sol: Incident voltage, $V = 6000$ V

$$\text{Incident current, } I = \frac{V}{Z_L} = \frac{6000}{500} = 12 \text{ A}$$



Time taken to reach 600 km

$$T = \frac{600}{3 \times 10^5} = 2 \text{ msec}$$

After $4T$ are 0.008s voltage and currents at sending end side are

$$V_s = V = 6000 \text{ V}$$

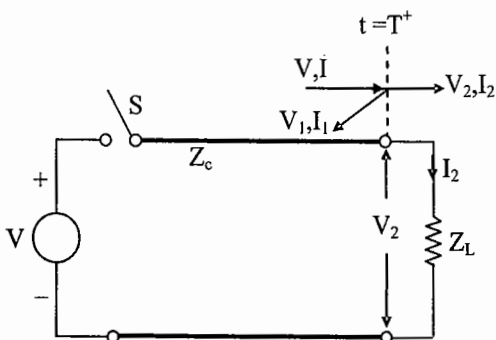
$$I_s = 5I = 60 \text{ A}$$

Electrical equivalent circuit for the calculation of refracted quantities at discontinuity

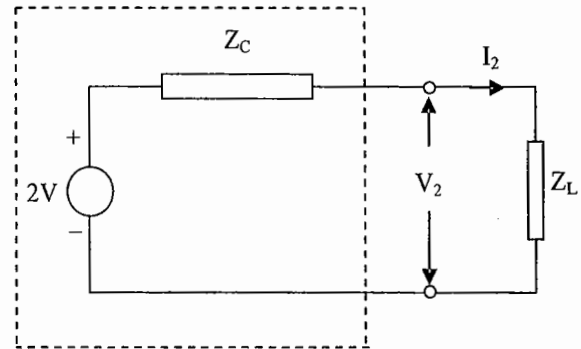
Incident waves are V, I

Reflected waves are V_1, I_1

Refracted waves are V_2, I_2



The lumped electrical equivalent circuit to find refracted quantities at T^+ is as follows



Equivalent circuit

Reflected $(V_1, I_1) = \text{refracted } (V_2, I_2) - \text{incident } (V, I)$

- Refracted voltage $V_2 = 2V \frac{Z_L}{Z_L + Z_c}$

Coefficient of voltage refraction

$$V_{\text{refraction}} = \frac{V_2}{V} = \frac{2Z_L}{Z_L + Z_c}$$

- Reflected voltage $V_1 = V_2 - V$

$$= 2V \frac{Z_L}{Z_L + Z_c} - V$$

$$= V \left[\frac{2Z_L}{Z_L + Z_c} - 1 \right]$$

$$V_1 = V \left[\frac{Z_L - Z_c}{Z_L + Z_c} \right]$$

Coefficient of voltage reflection

$$V_{\text{reflection}} = \frac{V_1}{V} = \left[\frac{Z_L - Z_c}{Z_L + Z_c} \right]$$

- Refracted current $I_2 = \frac{2V}{Z_L + Z_c}$

$$I_2 = 2I \frac{Z_c}{Z_L + Z_c}$$

Co-efficient of current refraction

$$i_{\text{refraction}} = \frac{I_2}{I} = \frac{2Z_c}{Z_L + Z_c}$$

- Reflected current $I_1 = I_2 - I$

$$I_1 = 2I \frac{Z_c}{Z_L + Z_c} - I$$



$$I_1 = I \left[\frac{Z_C - Z_L}{Z_C + Z_L} \right]$$

$$I_1 = -I \left[\frac{Z_L - Z_C}{Z_L + Z_C} \right]$$

Co-efficient of current reflection

$$i_{\text{reflection}} = \frac{I_1}{I} = - \left[\frac{Z_L - Z_C}{Z_L + Z_C} \right]$$

For open circuit line ($Z_L = \infty$) :

$$V_{\text{reflection}} = 2$$

$$V_{\text{reflection}} = 1$$

$$i_{\text{reflection}} = 0$$

$$i_{\text{reflection}} = -1$$

For short circuit line ($Z_L = 0$)

$$V_{\text{reflection}} = 0$$

$$V_{\text{reflection}} = -1$$

$$i_{\text{reflection}} = 2$$

$$i_{\text{reflection}} = 1$$

Example 2.5.2:

A traveling wave due to lightning with an incident voltage V travels through the overhead line of surge impedance of 400Ω and enters a cable of surge impedance of 40Ω . What is the voltage entering the cable at the junction? (IES-06)

- (a) $V / 11$ (b) $4V / 11$
(c) $2V / 11$ (d) V

Sol: Refracted voltage into the cable is,

$$V_2 = 2V \frac{Z_{\text{line}}}{Z_{\text{line}} + Z_{\text{cable}}}$$

$$V_2 = 2V \frac{40}{400 + 40}$$

$$V_2 = \frac{2V}{11}$$

Example 2.5.3:

Match List I (load) with List II (relationship) in respect of forward current i_f and the reflected current i_r at the receiving end of the transmission line. Select the correct answer using the codes given below the Lists: (IAS-07)

List I

- A. Open-circuit
B. Short-circuit
C. $R_L = Z_0$
D. $R_L < Z_0$

List II

1. $i_r = 0$
2. $i_r = -i_f$
3. Partial reflection
4. $i_r = i_f$

Codes:

	A	B	C	D
(a)	2	4	1	3
(b)	4	2	3	1
(c)	2	4	3	1
(d)	4	2	1	3

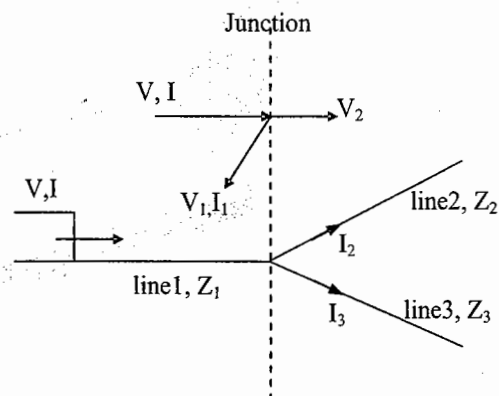
Sol: OC $\rightarrow i_r = -i_f$

SC $\rightarrow i_r = i_f$

$R_L = Z_0 \rightarrow i_r = 0$

$R_L < Z_0 \rightarrow$ Partial reflection

Bifurcated line (or) Forked line (or) Parallel line:

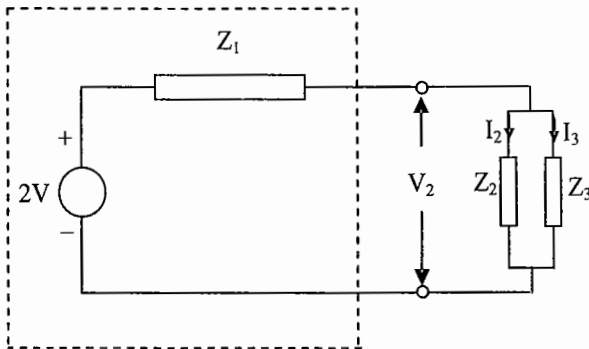


Incident waves are V, I

Reflected waves are V_1, I_1

Refracted waves are V_2, I_2, I_3

The electrical equivalent circuit at T^+ to find refracted quantities is as follows



Equivalent circuit

$$Z_{eq} = \frac{Z_2 Z_3}{Z_2 + Z_3}$$

$$\text{Refracted voltage } V_2 = 2V \frac{Z_{eq}}{Z_{eq} + Z_1}$$

$$V_2 = 2V \frac{Z_2 Z_3}{Z_2 Z_3 + Z_1 Z_2 + Z_1 Z_3}$$

Coefficient of voltage refraction is,

$$V_{\text{refraction}} \frac{V_2}{V_1} = \frac{2Z_2 Z_3}{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}$$

Reflected voltage $V_1 = V_2 - V$

$$= 2V \frac{Z_2 Z_3}{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1} - V$$

$$V_1 = V \left[\frac{Z_2 Z_3 - Z_1 Z_2 - Z_1 Z_3}{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1} \right]$$

Coefficient of voltage reflection

$$V_{\text{reflection}} = \frac{V_1}{V} = \frac{Z_2 Z_3 - Z_1 Z_2 - Z_1 Z_3}{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}$$

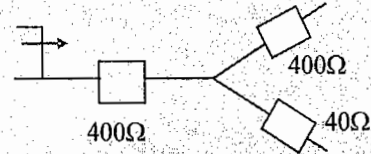
Refracted currents

$$I_2 = \frac{V_2}{Z_2} = I_3 = \frac{V_2}{Z_3}$$

Reflected current

$$I_1 = (I_2 + I_3) - I$$

Example 2.5.4:



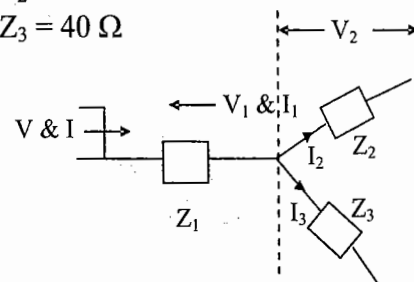
The magnitude of voltage and current which enter the bifurcated lines .

Sol: Surge voltage = 20 KV

$$Z_1 = 400 \Omega$$

$$Z_2 = 400 \Omega$$

$$Z_3 = 40 \Omega$$



$$V_2 = 2V \left[\frac{Z_2 Z_3}{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1} \right] \text{ kV}$$

$$= 2 \times 20 \text{KV} \left[\frac{400 \times 40}{(400 \times 400) + (400 \times 40) + (40 \times 400)} \right]$$

$$V_2 = 3.333 \text{ KV}$$

$$I_2 = \frac{V_2}{Z_2} = \frac{3.33 \times 1000}{400} = 8.33 \text{ A}$$

$$I_3 = \frac{V_2}{Z_3} = \frac{3.333 \times 1000}{40} = 83.33 \text{ A}$$

Example 2.5.5:

What is the first impressed surge voltage and currents into line-3 from following network.



$$Z_1 = 400 \Omega \quad Z_2 = 300 \Omega \quad Z_3 = 250 \Omega$$

$$(a) 38.96 \text{ kV}, 155.8 \text{ kA}$$

$$(b) 155.84 \text{ kV}, 623.4 \text{ A}$$

$$(c) 77.92 \text{ kV}, 31.17 \text{ A}$$

$$(d) 311.68 \text{ kV}, 1246.8 \text{ A}$$